Arunabha Majumder

Research Engineer | Wearable Robotics, Exoskeletons, Actuators & Experimental Validation

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SUMMARY

I design and build exoskeletons. In my PhD at Aalborg University I took a shoulder exoskeleton from hand sketches to a tested prototype: compliant actuator design, SolidWorks, 3D printing, motor and sensor integration, control in MATLAB and Python, and experiments with 12 participants. The design cut energy use by more than 25% and won the Best Paper Award at IFToMM ISRM 2026. I have also worked hands-on with soft pneumatic artificial muscles.

KEY RESULTS

- **>25%** energy reduction with a hybrid-actuated shoulder exoskeleton
- **>15%** lower user muscle effort with an Assist-as-Needed controller built in a team of 3
- **<10%** RMS force-estimation error for physical human-robot interaction

RESEARCH EXPERIENCE

Research Engineer / PhD Researcher – Wearable Robotics & Exoskeletons (VIEXO Project) | 2023 – Present
Aalborg University, Department of Materials and Production, Aalborg, Denmark

- Reduced energy consumption by more than 25% (Best Paper Award, IFToMM ISRM 2026) by designing a hybrid-actuated shoulder exoskeleton that combines a compliant parallel spring with a 6 Nm motor in a new actuator and transmission concept.
- Developed functional robot prototypes from concept to experimental validation through iterative design in SolidWorks and 3D-printed prototypes.
- Integrated the motor, encoder and force sensors on the exoskeleton, and worked in a team of 3 on its Assist-as-Needed controller: I helped with the equations, such as the Fourier fit, and defined how the GUI should behave. User muscle effort dropped by more than 15%, measured with EMG.
- Achieved below 10% RMS force-estimation error, validated against a force sensor, by developing a force-sensing method for a 5-bar planar parallel robot using variable-stiffness actuators and the robot Jacobian; applied it to measure human arm impedance in the gait lab.
- Collected the full experimental dataset with EMG and motion capture in experiments with 12 participants across 3 assistance modes and 2 load conditions.
- Supervised Bachelor's (B.Tech) students and taught university courses (2023 – 2025); disseminated results at international conferences.

MTech Researcher – Pneumatic Artificial Muscles (Master's Thesis) | 2020 – 2022
CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India

- Brought position error down to 0.3-0.78%, better than a standard PID, by working on the outer PID loop of a neural-network gain-scheduled controller in MATLAB/Simulink. I collected the experimental data, fed it to the lab's existing neural network and used its output to set the PID gains (IEEE CONECCT 2022).
- Fabricated and assembled soft pneumatic artificial muscles to an existing lab design and tested them on the lab's setup with valves, pressure sensors and DAQ.

Bachelor's Capstone – Upper-Limb Rehabilitation Exoskeleton | 2018 – 2019
Siddaganga Institute of Technology, Tumakuru, India

- Led CAD design, machining, hands-on assembly and user testing of a rehabilitation exoskeleton; Best Major Project Award and INR 100,000 TATA Technologies Innovation Grant.

PUBLICATIONS

- Majumder, A., Wagner, J.W., Zhu, Y., Oliveira, A.S., and Bai, S. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." Robotics and Mechatronics: ISRM 2026, MMS, vol. 213, Springer, 2026. Best Research Paper Award.
- Majumder, A., Sarkar, D., Chakraborty, S., Singh, A., Roy, S.S., and Arora, A. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." IEEE CONECCT, 2022.

EDUCATION

Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present
Aalborg University, Department of Materials and Production, Aalborg, Denmark

M.Tech (Master's), Mechatronics | 2020 – 2022
AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

B.Tech, Mechanical Engineering | 2015 – 2019
Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India

SKILLS

- **Design & prototyping:** SolidWorks (CSWA), 3D printing; machining and assembly (Bachelor's)
- **Programming:** MATLAB/Simulink, Python, LabVIEW (basic), Arduino
- **Control:** Assist-as-Needed control (team work: equations incl. Fourier fit, GUI logic), PID tuning, kinematics and Jacobian-based force estimation
- **Sensors & actuators:** Motors, encoders, force/torque sensors, pressure sensors, DAQ, EMG, Qualisys motion capture
- **Drawing:** Diploma in Fine Arts; hand sketches of concepts, technical illustrations and figures for papers, posters and demos
- **Languages:** English (C1, fluent), Bengali (Native), Hindi (Conversational)

AWARDS

- Best Research Paper Award, 9th IFToMM International Symposium on Robotics and Mechatronics (2026)
- Certified SolidWorks Associate (CSWA); TATA Technologies Innovation Grant – INR 100,000 (2019); Best Major Project Award (2019)

REFERENCES

Available upon request.